MIL-S-25392 B(ASG)
8 MAY 1968
Superseding
MIL-S-25392A(ASG)
20 December 1963

MILITARY SPECIFICATION

SANDWICH CONSTRUCTION, PLASTIC RESIN, GLASS FABRIC BASE, LAMINATED FACINGS AND URETHANE FOAMED-IN-PLACE CORE, FOR STRUCTURAL APPLICATIONS

This specification has been approved by the Department of the Air Force and by the Naval Air Systems Command.

1. SCOPE

1.1 Scope. - This specification covers plastic molded sandwich construction using glass fabric base plastic laminate facings and urethane foamed-in-place core material for aircraft structural applications up to 350° F, including all aircraft external plastic parts, and other weapons applications.

1.2 Classification. - The molded sandwich construction, foamed-in-place core material shall be of the following types and classes, as specified (see 6.2):

Type I	General purpose - for exposure to 165° F
Type II	Heat resistant - for exposure to 350° F
Class 1	Radar purpose (electrical) (see 3.5)
Class 2	General purpose (nonelectrical)

2. APPLICABLE DOCUMENTS

2.1 The following documents, of the issue in effect on date of invitation for bids or request for proposal, form a part of this specification to the extent specified herein:

SPECIFICATIONS

Federal

L-P-383	Plastic Material, Polyester Resin, Glass Fiber Base, Low Pressure Laminated

Military

MIL-R-7575	Resin, Polyester, Low-Pressure Laminating
MIL-C-8087	Core Material, Foamed-in-Place, Urethane Type
MIL-C-9084	Cloth, Glass, Finished, for Polyester Resin Laminates
MIL-R-9300	Resin, Epoxy, Low Pressure Laminating

FSC 9330

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MIL-P-9400	Plastic Laminate Materials and Sandwich Construction, Glass Fiber Base, Low Pressure Aircraft Structural, Process Specification Requirements
MIL-R-25042	Resin, Polyester, High Temperature Resistant, Low Pressure Laminating

STANDARDS

Federal

FED. TEST METHOD STD. NO. 406	Plastics: Methods of Testing

Military

MIL-STD-143	Specifications and Standards, Order of Precedence for the Selection of
MIL-STD-401	Sandwich Construction and Core Materials, General Test Methods

(Copies of specifications, standards, drawings, and publications required by suppliers in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

3. REQUIREMENTS

3.1 Preproduction. - This specification provides for preproduction inspection.

3.2 Materials. - Materials other than those specified may be used, provided they are suitable for the purpose. Specifications governing these materials shall be selected in accordance with MIL-STD-143.

3.2.1 Sandwich construction. - The sandwich construction shall consist of foamed-in-place core material bonded to faces of glass fabric base laminated plastic material.

3.2.2 Face material. - The face material shall conform to the workmanship requirements of the applicable plastic materials specifications for each type and class of sandwich construction.

3.2.2.1 Glass fabric. - The glass fabric shall conform to MIL-C-9084.

3.2.2.2 Laminating resins. - For type I applications, the laminating resin shall conform to MIL-R-7575 or MIL-R-9300. For type II applications, the laminating resin shall conform to MIL-R-9300 or MIL-R-25042. For class 1, type I and type II applications, the laminating resin shall also meet the additional electrical requirements of the applicable specification for the laminating resin.

3.2.3 Core material. - The core material shall conform to the corresponding type and class requirements of MIL-C-8087 for the particular type and class sandwich structure fabricated.

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3.2.4 Bonding. - The foamed-in-place core material shall be capable of bonding to the premolded plastic resin, glass fabric base low pressure laminated skins. The surface of the skin to be in contact with the foam shall be properly prepared to remove surface gloss and parting agents. An adhesive primer may be applied to the inner surface of the laminated skins before foaming to obtain a good bond. For class 1 materials, any adhesive primer used shall not impair the electrical properties specified in 3.5.

3.2.5 Sandwich weight. - The sandwich weight for given core densities shall not exceed the maximum allowable requirements of figure 1 for the test panel of dimensions described in 4.3.1.

3.3 Process. - A detailed description of the manufacturing and fabricating process and methods of control of manufacturing variables shall be submitted by the contractor in the form of a titled, numbered, and dated process specification, together with the number of panels specified in 4.3.1, and 4.3.2 (The process specification shall contain the results of all physical and mechanical tests required under preproduction testing) to the procuring activity for approval in conformance to MIL-P-9400 and this specification. After approval, the process specification shall form a part of this specification, and copies shall be made available by the contractor for use by authorized Government inspectors at the contractor's plant.

3.4 Mechanical strength properties. - The mechanical strength properties of the test specimens cut from sandwich panels fabricated as described in 4.4.1 shall be determined on the minimum and maximum sandwich weight (on a lb/cu.ft. basis, figure 1) that the manufacturer anticipates producing.

3.4.1 Mechanical strength properties at room temperature. - When tested in sandwich form, the mechanical strength prpperties of type I and type II materials shall conform to the requirements of figures 2 and 3, respectively.

3.4.2 Mechanical strength properties at 165° F. - The flexural, flatwise compression, and flatwise tensile strengths of the plastic sandwich, when tested at 165° F, shall be not less than 70 percent of the room temperature requirements as given in figure 1 for type I materials.

3.4.3 Mechanical strength properties at 350° F. - The flexural strength, flatwise compression, and flatwise tensile strengths for type II sandwich, when tested at 350° F, shall be not less than 75 percent of the room temperature requirements as given in figure 3.

3.5 Electrical properties for types I and II (see figure 4). - The electrical power transmission efficiency of foam sandwich panels, when tested at a frequency between 8.5 to 10 kilomegacycles (designed for optimum performance at 9,375 ±10 megacycles) and oriented at a 20-degree angle of incidence for both parallel and perpendicular polarization, shall be at least 95 percent in the "as received" condition and at least 90 percent after environmental conditioning (4.5.4) and testing in accordance with 4.6.7. No electrical tests are required for class 2 material.

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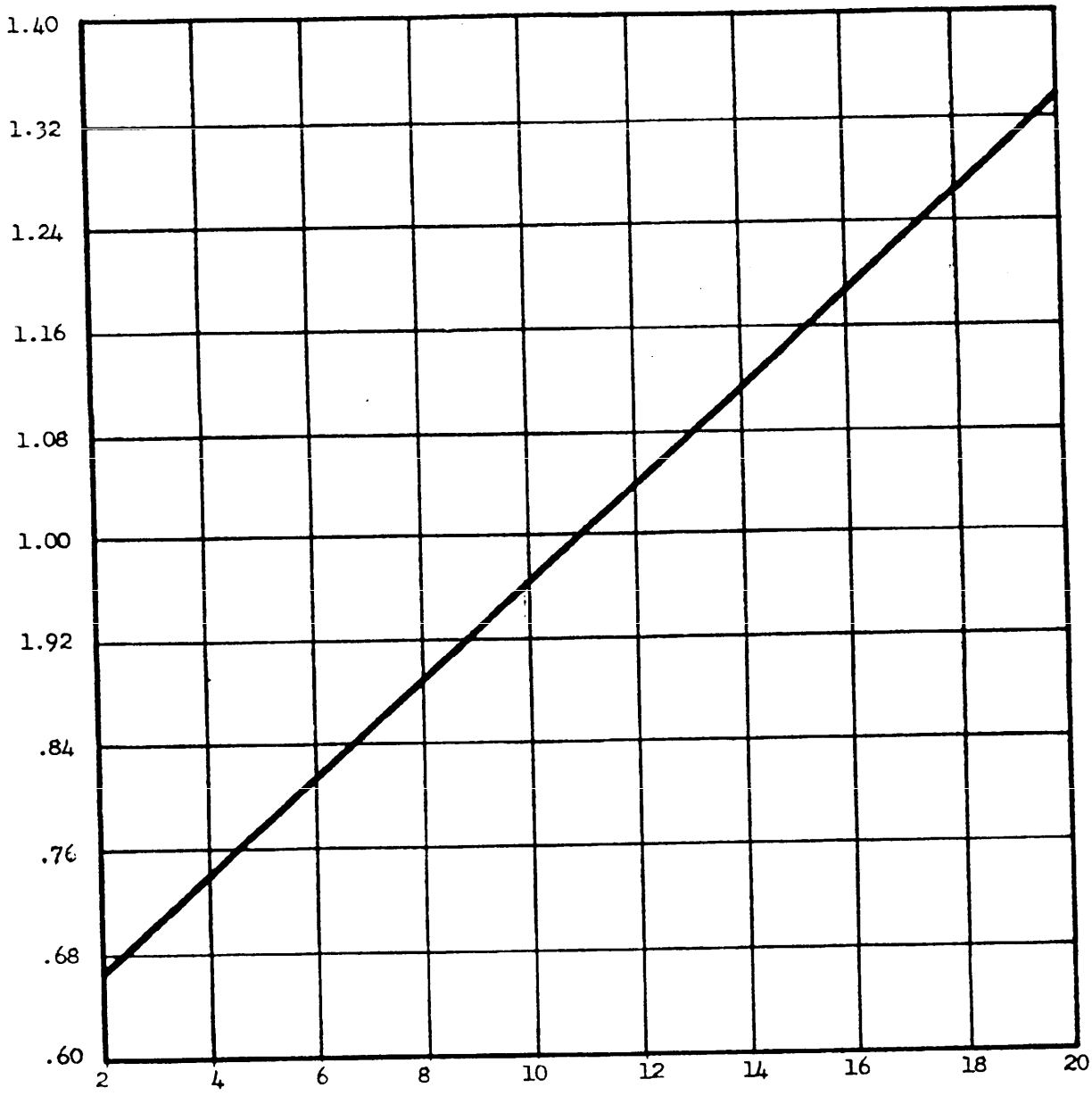


FIGURE 1. Maximum sandwich weight of foam panel versus core density

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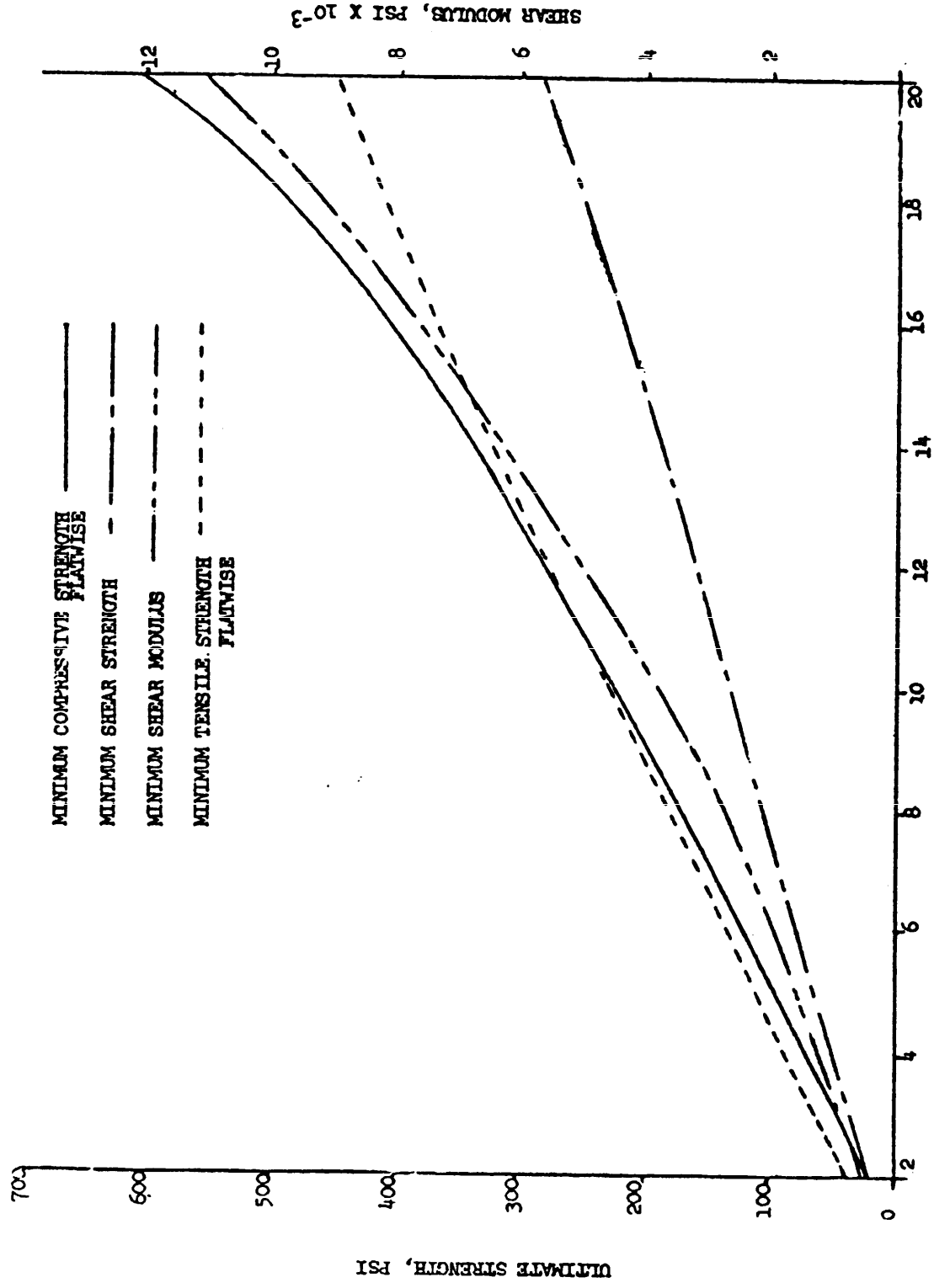


FIGURE 2. Mechanical strength properties for type I material

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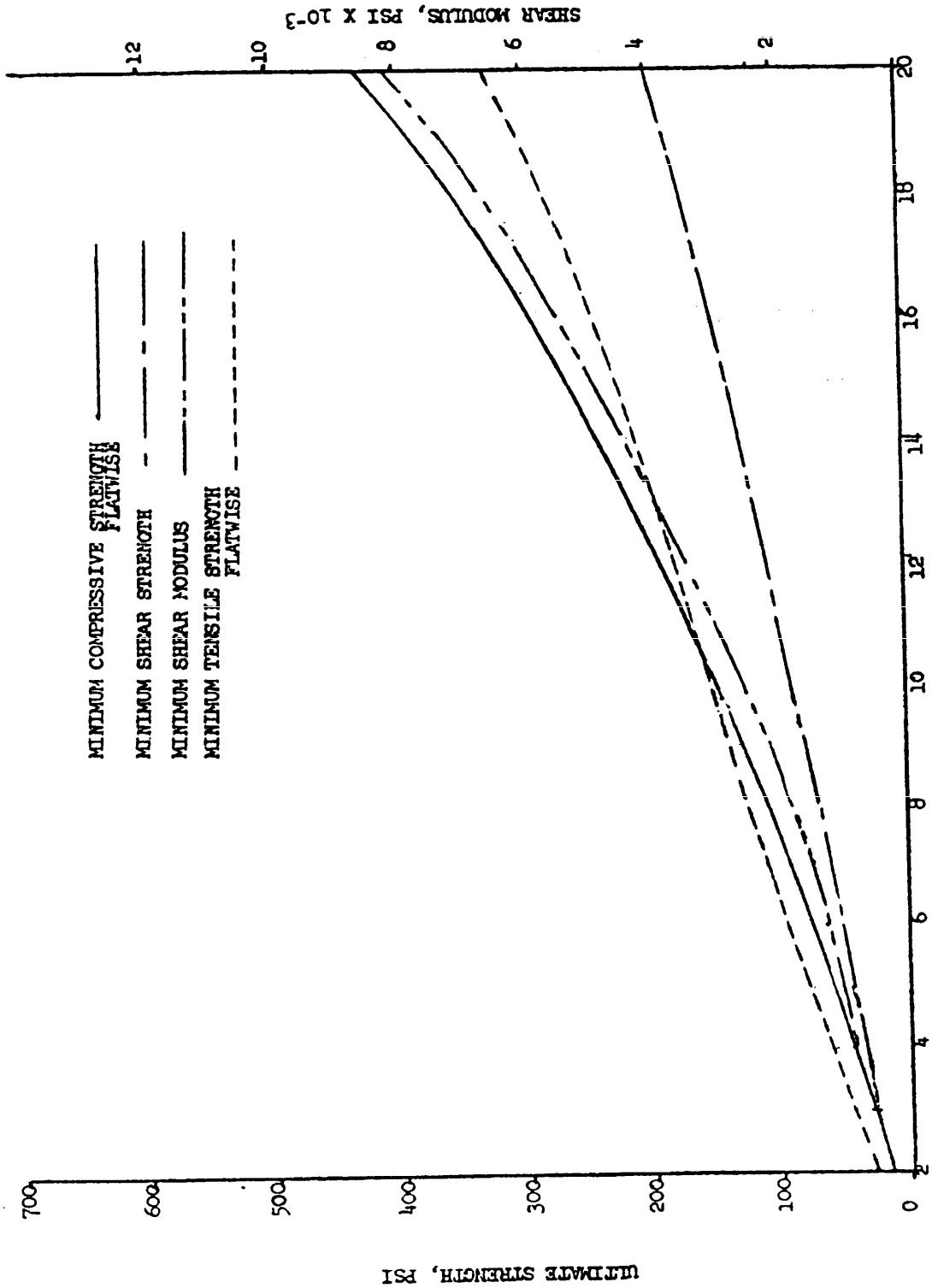


FIGURE 3. Mechanical strength properties for type II material

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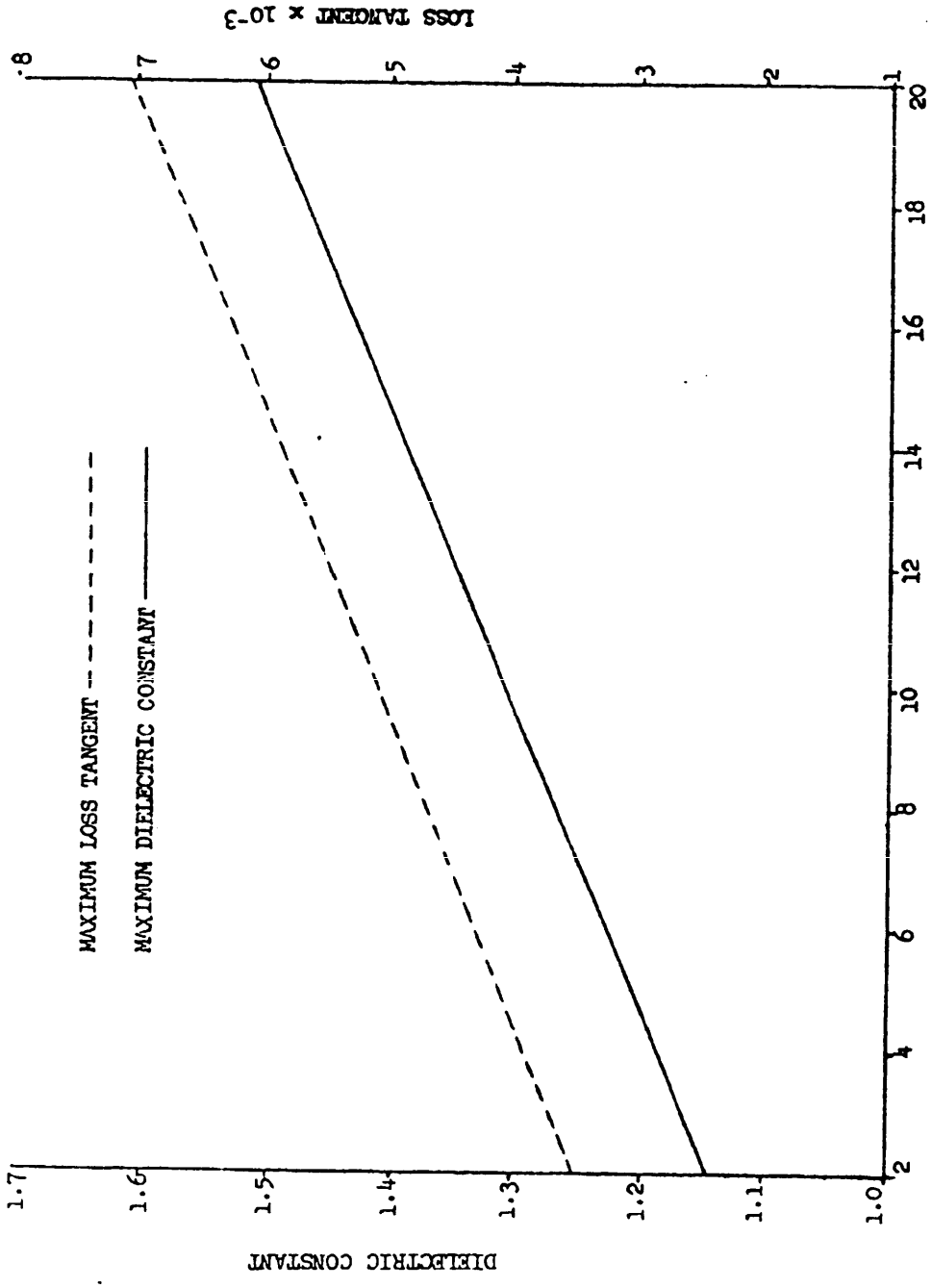


FIGURE 4. Electrical properties of class 1, types I and II materials (frequency - 8.5 to 10 KMC)

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3.6 Dimensions and weight. - The face thickness, overall sandwich construction thickness, and maximum weight in pounds per square foot, shall be as specified in the specifications, drawings, or contracts for the molded sandwich construction parts.

3.7 Workmanship. -

3.7.1 Cured foam core. - Workmanship requirements of MIL-C-8087 shall apply to the cured foamed-in-place core material used in sandwich construction.

3.7.2 Defects. - The minimum requirements for the sandwich and maximum allowable and reparable defects shall be completely described in the manufacturer's process specification. Any sandwiches having defects which exceed the maximum allowable defects in nature, size, or extent beyond satisfactory reparable condition shall be rejected. In general, the workmanship and quality of the plastic sandwich part shall conform to MIL-P-9400 and this specification.

3.7.3 Blistering. - The finished sandwich part shall show no blistering under service or service simulated conditions. Drilling of holes on the inner skin of the finished part or other methods of inducing porosity on the inner skin to insure against blistering under service or service simulated conditions is prohibited.

3.8 Dimensions and makeup of sandwich construction. - The thickness, dimensions, and makeup of the laminates and core material, comprising the number of plies of the various fabrics, their position and direction, type of resin and other specific information for the sandwich construction shall be specified in drawings, specifications,or contracts for the parts.

3.9 Overlays. - Glass fabric base low pressure laminated plastic material shall not be made with a gel resin overlay, integrally molded or otherwise, or with any other such thick resin surface, or any other surfacing material, unless such overlay or surfacing is specifically approved by the procuring activity.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. - Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or order, the supplier may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

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4.2 Classification of inspections. - The inspection of the sandwich construction shall be classified as follows:

(a) Preproduction inspection (4.3)
(b) Quality conformance inspection (4.4)

4.3 Preproduction inspection. -

4.3.1 Preproduction sample. - Prior to production of parts on contract, a sample flat sheet of sandwich construction shall be made up in accordance with the approved process specification of each of the types and classes of core materials to be employed in production parts and shall be tested for the applicable properties specified in figures 2 and 3. In the event that parts having different core densities are produced, two panels each of the minimum and maximum core density used shall be submitted. Flat panels 1/2-inch overall thickness shall be foamed in a vertical direction between MIL-C-9084, type VIII glass-fabric base parallel laminated skins, 0.030 ±0.003 inch thick, conforming to the workmanship requirements of L-P-383. The test panel shall be marked to show the direction of foaming. The method used for mixing the components, as well as for pouring and curing, shall be in accordance with the manufacturer's instruction sheet. Specimens from these panels shall be tested to determine conformance with the applicable requirements specified herein. These tests need be made only once and need not be repeated for new orders or different parts provided that the materials and processes have not been changed, and that a certified statement to this effect is furnished the inspector with a copy of the report of the tests made. However, in those cases where the Government inspector considers it necessary in order to obtain material meeting the requirements of this specification, he shall have the right to require the performance of any of the tests at any time. More than one sheet may be made and used for the tests provided that they are all comparable as to material and process of fabrication.

4.3.2 Electrical tests - class 1 materials. - Electrical tests on class 1 materials shall be performed by the procuring service on two foam sandwich panels, one each of the minimum density core material to be produced by the manufacturer. The foam sandwich panels shall be 2 feet by 3 feet in overall size, with the same construction specified in 4.3.1, except that the thickness shall be determined by the manufacturer on the basis of electrical design characteristics in order to conform to the requirements specified in 3.5. The outside edge of the panels shall have a 1-inch wide solid laminate border of the same type laminate used for the skin (see 6.4). The core densities and fabrication process (3.3) of the panels shall be the same as those for the panels tested for mechanical strength properties (3.4).

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4.4 Quality conformance inspection. - Quality conformance inspection shall consist of the following:

(a)	Visual examination	(4.6.1)
(b)	Weight	(4.6.2)
(c)	Compression flatwise	(4.6.3)

4.4.1 Sampling. - Unless otherwise specified by the procuring activity, from each lot of foamed-in-place core material and glass fabric laminate to be used in the fabrication of parts, at least one sample flat panel of sandwich construction plastic material shall be made up in accordance with the approved process specification of each of the types of core materials to be employed in production. This panel shall be tested for flatwise compressive strength at room temperature, and 165° F for type I and 350° F for type II in accordance with 4.6.3, 4.6.4, and under test conditions of 4.5. The overall requirements (density, etc.) shall conform to figures 2 and 3. Five specimens shall be tested for each test from each sample.

4.4.2 Lot. - A lot of foamed-in-place core material or glass fabric base plastic laminate material shall be defined as all foamed-in-place core material or glass fabric base plastic laminate material representative of the same manufacturer's batch or shipment.

4.4.3 Rejection and retest. - In case of failure of the material to pass the specified tests, an additional sample representative of the same lot shall be tested. If this sample fails the specified tests, the material shall not be used in the production of parts.

4.5 Test conditions. -

4.5.1 Room temperature conditions. - Tests may be performed in an atmosphere where the temperature ranges from 65° to 80° F and relative humidity from 30 to 70 percent. However, in case of any question as to the validity of test results, test conditions shall be 73° ±2° F and 50 ±4 percent relative humidity. Specimens shall be tested immediately after being exposed for 4 days to the room temperature conditions.

4.5.2 Exposure at 165° F. - Specimens shall be exposed for 30 minutes at a temperature of 165° ±4° F in a circulating air test chamber previously heated to the temperature of test. Immediately after the 30-minute exposure, the compression and tensile specimens shall be tested at the same temperature (165° ±4° F). The flexure specimens shall be tested at room temperature immediately after the 30-minute exposure at 165° ±4° F.

4.5.3 Exposure at 350° F. - Specimens shall be exposed for 30 minutes at a temperature of 350° ±5° F in a circulating air test chamber previously heated to the temperature of the test. Immediately after the 30-minute exposure, the compression and tensile specimens shall be tested at the same temperature (350° ±5° F). The flexure specimens shall be tested at room temperature immediately after the 30-minute exposure at 350° ±5° F.

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4.5.4 Environmental conditioning for electrical test panels. - Either a thermometer or thermocouple shall be inserted into the approximate centerline of the specimen. The specimen shall be heated or cooled to temperature equilibrium, 75° C (165° F) and -54° C (-65° F), respectively. The specimen shall be exposed at temperature equilibrium for 5 minutes. One cycle shall consist of 5 minutes at 75° C and 5 minutes at -23° C and the number of complete cycles shall be 25. The specimen shall then withstand simulated altitude changes of 50 cycles, holding 1 minute each at 8,000 and 50,000 feet.

4.6 Inspection methods. -

4.6.1 Visual examination. - All face and core materials shall be visually inspected for compliance with the requirements specified in 3.2.2 and 3.2.3. The requirements for plastic cover sheets shall be in accordance with Fed. Test Method Std. No, 406. Each part molded of the plastic materials shall be inspected during and after manufacture to determine compliance with the requirements of this specification and with the requirements specified herein for workmanship, foreign materials, gaps and laps, overlays, and design and dimensions. All the inspections shall be made before any coating or finish of any kind is applied to the part after molding.

4.6.2 Weight. - Sandwich weight shall be determined on samples (minimum of 1 square foot) from the weight and dimensions by calculating the area and dividing the weight of the piece in pounds by the area. Weight shall be determined to an accuracy of at least 1 percent.

4.6.3 Compression flatwise. - Flatwise compressive strength shall be determined on sandwich specimens of foamed-in-place core material 2 inches by 2 inches by approximately 1/2 inch in dimension, carefully machined from the test panel with the 2-by-2-inch surface being perpendicular to the direction of loading. The dimensions of the specimens shall be measured to an accuracy of at least 0.5 percent. Ten specimens shall be tested as machined and the results averaged. The specimens shall be loaded between flat plates at a uniform rate of head travel of not more than 0.05 inch per minute. The compressive strength shall be reported as the value in pounds per square inch at 10 percent total compression or ultimate (maximum) strength, whichever is the lower (see 6.3). The amount of compression in inches shall be determined by measuring the movement of the loading head with a dial gage or equivalent. Head travel should be continuous and uniform and so adjusted that failure occurs within 3 minutes after beginning the test.

4.6.4 Flatwise tensile test. - Tensile strength flatwise shall be determined in accordance with the method designated as sandwich tension in MIL-STD-401. Ten specimens shall be tested and the results averaged.

4.6.5 Thickness. - The sandwich panel thickness shall be taken as the average of 20 randomly distributed readings each measured to the nearest 0.001 inch. The face thicknesses shall be measured to the nearest 0.001 inch by taking several measurements on the premolded faces before fabrication of the sandwich panel.

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4.6.6 Electrical properties. - The electrical tests shall be performed by the shunted line waveguide method.

4.7 Packaging, packing, and marking. - Preparation for delivery shall be examined for conformance to section 5.

5. PREPARATION FOR DELIVERY

5.1 Application. - The packaging, packing, and marking requirements shall be as specified in the specifications, drawings, or contracts for the parts fabricated from the plastic material.

6. NOTES

6.1 Intended use. - The foamed-in-place core molded plastic sandwich construction is intended for use in radomes and structural parts of aircraft and weapons items. (All exterior aircraft plastic parts are considered structural parts.)

6.2 Ordering data. - Procurement documents should specify:

(a) Title, number, and date of this specification.
(b) Overall thickness and thickness of face material; makeup face sheet thickness as to number of plies and layup, and identification of the plies as to fabric number under MIL-C-9084.
(c) Fabric finish, type of laminating resin under the appropriate specification for the resin; and type, class, and dimensions of core under MIL-C-8087. Type and class (see 1.2).
(d) Packaging, packing, and marking requirements (see 5.1).

6.3 Definition of compressive strength (see 4.6.4). - Specifying strength to be reported as to the value at a particular percent of total compression shall not be interpreted to mean the same as conventional offset methods such as are used for determining yield stress in a ductile metal. For purposes of this specification, the interpretation shall be that strengths reported on a total compression basis shall be the value which would be fixed at the intersection of the stress-strain curve and a line perpendicular to the horizontal axis at the specified strain value (10-percent deformation as determined by head travel).

6.4 The purpose of a 1-inch-wide solid laminate border of the same laminate used for the skin is to minimize moisture transmission through the panel edged during environmental conditioning.

Custodians:
Navy - AS
Air Force - 11

Reviewer activities:
Navy - AS
Air Force - 11, 84

Preparing activity:
Navy - AS

Project No. 9330-N325

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SPECIFICATION ANALYSIS SHEET

Form Approved Budget Bureau No. 119-R004

INSTRUCTIONS

This sheet is to be filled out by personnel either Government or contractor, involved in the use of the specification in procurement of products for ultimate use by the Department of Defense. This sheet is provided for obtaining information on the use of this specification which will insure that suitable products can be procured with a minimum amount of delay and at the least cost. Comments and the return of this form will be appreciated. Fold on lines on reverse side, staple in corner, and send to preparing activity.

SPECIFICATION

MIL-S-25392B(ASG) Sandwich Constr., Platic Resin, etc., Structural Applic.

ORGANIZATION | CITY AND STATE

CONTRACT NO. | QUANTITY OF ITEMS PROCURED | DOLLAR AMOUNT $

MATERIAL PROCURED UNDER A

( ) Direct Government Contract ( ) Subcontract

1. HAS ANY PART OF THE SPECIFICATION CREATED PROBLEMS OR REQUIRED INTERPRETATION IN PROCUREMENT USE?

A. GIVE PARAGRAPH NUMBER AND WORDING.

B. RECOMMENDATIONS FOR CORRECTING THE DEFICIENCIES.

2. COMMENTS ON ANY SPECIFICATION REQUIREMENT CONSIDERED TOO RIGID.

3. IS THE SPECIFICATION RESTRICTIVE?

( ) YES ( ) NO IF "YES", IN WHAT WAY?

4. REMARKS (Attach any pertinent data which may be of use in improving this specification. If there are additional papers, attach to form and place both in an envelope addressed to preparing activity.)

SUBMITTED BY (Printed or typed name and activity) DATE

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